MASTER PREPARATION RESOURCE · 52 AUDITED QUESTIONS

T2D Glucose Digital Twin — Jury Q&A Bank

Comprehensive spoken answers, one-sentence summaries, exact source artifacts, follow-up defenses, and negative result disclosures for Akshaya R and Nikhil VS.

Presenters: Akshaya R (BGSIT) & Nikhil VS (EWIT) · **Live Site:** `t2d-glucose-digital-twin.akshay.blitz.cloud` · **Default Branch:** `release/candidate-clean` (SHA `1c6e51b4`)

Category 1: Problem and Healthcare Need

Q01 What exact clinical problem does this project solve?

Direct Spoken Answer: Commercial continuous glucose monitors only sound alarms *after* blood glucose has already crossed 180 mg/dL, which is too late to prevent the excursion. Our system predicts the impending rise 30 to 60 minutes in advance, giving patients an actionable window for light activity—such as a 15-minute walk—or meal moderation before the spike peaks.

> ***One-Sentence:*** *We replace reactive post-spike CGM alarms with a 30- to 60-minute forward forecast to enable non-pharmacological behavioral interventions in type 2 diabetes.*

`Source: Root README.md & docs/jury/JURY_PREPARATION_GUIDE.md §2`

Likely Follow-Up: Why is a 30-minute warning useful in type 2 diabetes when patients don't take fast-acting insulin?

In type 2 diabetes, light exercise stimulates GLUT4 glucose transporter translocation directly into skeletal muscle independent of insulin, but this clearance mechanism is most effective before or during the initial glucose upslope, not after severe hyperglycemia has set in.

⚠️ What NOT to Claim: Do not claim the system currently improves HbA1c, prevents hospitalizations, or has undergone clinical trials.

Q02 What is a continuous glucose monitor (CGM), and what does it measure?

Direct Spoken Answer: A continuous glucose monitor is a wearable sensor that measures glucose in the interstitial fluid between subcutaneous cells every 5 minutes. It does not measure blood directly; interstitial glucose lags capillary blood glucose by 5 to 15 minutes due to physiological diffusion time.

> ***One-Sentence:*** *It is a wearable subcutaneous sensor measuring interstitial fluid glucose every 5 minutes, which has a 5- to 15-minute physiological lag behind capillary blood.*

`Source: backend/input_contract.py & docs/evidence/real_data/PREREGISTRATION.md §2`

Likely Follow-Up: How does your model account for this 5- to 15-minute interstitial lag?

The Unscented Kalman Filter tracks glucose clearance kinetics explicitly, while the PR-TFT network observes 24 hours of historical readings, allowing it to learn the temporal rate of change.

⚠️ What NOT to Claim: Do not claim interstitial readings are instantaneous or identical to venous blood glucose.

Q03 What exactly is an "adverse glucose spike" in your system?

Direct Spoken Answer: In our preregistered protocol, an adverse spike is defined as interstitial glucose crossing from below 180 mg/dL to 180 mg/dL or higher, preceded by a rise of at least 50 mg/dL in the prior 60 minutes, and persisting above 180 mg/dL for at least 15 minutes (3 consecutive readings).

> ***One-Sentence:*** *A glucose rise crossing 180 mg/dL with a 50 mg/dL jump in the preceding hour that persists for at least 15 minutes.*

`Source: docs/evidence/real_data/PREREGISTRATION.md §3 & backend/train_pr_tft.py`

Likely Follow-Up: Why require a 15-minute persistence rule?

Sensor noise or temporary physical compression on the wearable can produce transient single-sample spikes; requiring 3 consecutive 5-minute readings ensures we only evaluate genuine metabolic excursions.

⚠️ What NOT to Claim: Do not claim that 180 mg/dL is an absolute diagnostic threshold for every individual; it is the standard international Time-in-Range ceiling.

Q04 Why can't a patient just use the standard high-glucose alarm on their existing CGM?

Direct Spoken Answer: Standard CGM alarms are purely reactive; they trigger only after the threshold is breached. Furthermore, if you set the alarm threshold lower—say at 150 mg/dL—to catch spikes early, you get overwhelming false alarms during normal postprandial fluctuations. Our model evaluates rate of change, history, and physiological state to predict whether a crossing will occur.

> ***One-Sentence:*** *Standard alarms trigger after the threshold is crossed, and lowering their threshold creates intolerable false alarms during normal digestion.*

`Source: docs/demo/JURY_BRIEF.md §5 & docs/presentation/presentation.html Slide 2`

Likely Follow-Up: Did your system actually beat the CGM's built-in trend arrows?

In our locked real-patient test, a simple 30-minute linear trend caught 44.4% of spikes but fired 3.84 false alarms per day—nearly 4 false alarms daily. Our hybrid model controlled false alerts to 0.11 per day with 65.5% precision, eliminating alert fatigue.

⚠️ What NOT to Claim: Do not claim that our model caught more spikes than the linear trend on real patients; it caught fewer (6.9%), but with far higher precision and acceptable false-alarm burden.

Category 2: What Makes It a Digital Twin

Q05 Everyone claims to build a "digital twin." What makes yours a digital twin rather than just a machine learning model?

Direct Spoken Answer: A pure machine learning model takes inputs and outputs numbers from a black box. Our system maintains an explicit computational state representation of the individual patient: their static EHR priors and a recursive Unscented Kalman Filter tracking biophysical states—glucose, remote insulin action, and gut carbohydrate absorption compartments. The neural network only predicts the residual correction over this biophysical twin.

> ***One-Sentence:*** *It maintains an internal, continuously synchronized biophysical state model (Bergman ODEs via UKF) that produces a mechanistic baseline, which the neural network refines rather than replaces.*

`Source: backend/ukf_twin.py & docs/jury/JURY_PREPARATION_GUIDE.md §3`

Likely Follow-Up: Does your digital twin adapt its physiological parameters to each patient over time?

Currently, the UKF adapts dynamic states like insulin action and active gut carbs continuously with each 5-minute reading, while static parameters use EHR-derived priors. Full nightly Maximum A Posteriori (MAP) parameter fitting is documented in our blueprint as future work.

⚠️ What NOT to Claim: Do not claim that the model has performed nightly MAP fitting on individual patients in the deployed prototype; the TwinCard explicitly marks interval calibration as prior-based.

Q06 What data does the digital twin actually synchronize with?

Direct Spoken Answer: It synchronizes static EHR context—age, sex, BMI, baseline HbA1c, eGFR, diabetes duration, and medication classes—with streaming wearable telemetry: 5-minute CGM readings, heart rate, heart rate variability (RMSSD), steps, sleep stage, and logged carbohydrate intake.

> ***One-Sentence:*** *Static EHR clinical context combined with 5-minute streaming CGM, heart rate, HRV, steps, sleep, and meal logs.*

`Source: backend/input_contract.py::REALISTIC_EHR_COLUMNS & backend/train_pr_tft.py`

Likely Follow-Up: What happens if a wearable doesn't provide heart rate or HRV?

In our reduced-input real-patient evaluation, missing wearable channels were masked out, and the model relied strictly on CGM, age, BMI, and HbA1c without inventing synthetic sensor values.

⚠️ What NOT to Claim: Do not claim that commercial consumer smartwatches reliably stream all these channels into hospital EHRs today.

Q07 Is the live prototype running on real patient streams or synthetic data?

Direct Spoken Answer: The live deployed prototype runs on a synthetic cohort of 249 served subjects. We strictly partitioned the cohort into 205 training subjects, 44 validation subjects, and 44 sealed test subjects that are never served. Our real-patient research was conducted separately in offline, locked evaluations on 161 clinical patients.

> ***One-Sentence:*** *The live workstation serves 249 synthetic subjects; real-patient evaluation was conducted offline under strict research protocols.*

`Source: backend/serve_pr_tft.py line 79 (SEALED_SPLITS) & docs/submission/CLAIM_EVIDENCE_LEDGER.md C06`

Likely Follow-Up: Why did you deploy synthetic subjects instead of real patient data?

Ethical privacy constraints and licensing agreements for clinical datasets (such as PhysioNet and Figshare) prohibit republishing identifiable or raw human physiological streams on a public cloud server.

> ⚠️ What NOT to Claim: Never claim the live web app is connected to live patients in a clinic.

Category 3: Model and Architecture

Q08 Explain the mathematical relationship between the physiological model and the neural network.

Direct Spoken Answer: The relationship is strictly additive: Final Forecast Median equals Twin Rollout Median plus Learned Residual Median. The biophysical model—a modified Bergman Minimal Model solved via UKF—projects a 2-hour trajectory assuming no unannounced meals. The Temporal Fusion Transformer observes 24 hours of history and predicts the residual delta (ΔG) across 7 quantiles.

> ***One-Sentence:*** *Final Forecast = UKF Biophysical Rollout + PR-TFT Learned Residual.*

`Source: backend/train_pr_tft.py::PRTFT.forward & docs/presentation/presentation.html Slide 5`

Likely Follow-Up: Why predict residuals instead of training the transformer directly on glucose?

Predicting residuals guarantees that when sensor data are noisy or during dropouts, the model is anchored by biophysical bounds rather than oscillating into unphysical negative glucose or exponential spikes.

⚠️ What NOT to Claim: Do not claim the learned residual has been individually validated as a specific biological mechanism; it is an empirical statistical adjustment.

Q09 What is the Unscented Kalman Filter (UKF), and why did you choose it over an Extended Kalman Filter (EKF)?

Direct Spoken Answer: The UKF is a recursive nonlinear state estimator. It uses deterministic sigma points to propagate mean and covariance through nonlinear differential equations without computing Jacobian matrices. We chose it over the EKF because insulin-glucose kinetics involve sharp nonlinearities (like gut saturation) where first-order Taylor approximations in the EKF often diverge.

> ***One-Sentence:*** *A nonlinear recursive filter that uses sigma points rather than Taylor series Jacobians, providing stable state tracking across nonlinear insulin-glucose dynamics.*

`Source: backend/ukf_twin.py & docs/jury/JURY_PREPARATION_GUIDE.md §4`

Likely Follow-Up: What states does the UKF track?

Plasma glucose G, remote insulin action X, plasma insulin proxy I, two gut glucose absorption compartments Q_1 and Q_2, insulin sensitivity S_I, and endogenous glucose production EGP.

⚠️ What NOT to Claim: Do not claim the UKF directly measures plasma insulin; it tracks a latent insulin action proxy driven by observed glucose dynamics and meal logs.

Q10 What is the PR-TFT architecture?

Direct Spoken Answer: PR-TFT stands for Physiological-Residual Temporal Fusion Transformer. It is an attention-based deep neural network featuring Variable Selection Networks to weight input relevance, Gated Residual Networks to skip unused features, and multi-head self-attention to capture long-range temporal dependencies across 24 hours of history.

> ***One-Sentence:*** *An attention-based architecture designed for multi-horizon time-series forecasting that outputs multi-quantile residual predictions over the biophysical twin.*

`Source: backend/train_pr_tft.py::PRTFT & runs/pr_tft_ablation_arm_d/config.json`

Likely Follow-Up: What are its key hyperparameters?

Hidden dimension of 96, 4 attention heads, 2 transformer layers, dropout of 0.1, trained with pinball quantile loss and binary cross-entropy hazard loss.

> ⚠️ What NOT to Claim: Do not claim to have invented the Temporal Fusion Transformer from scratch; we adapted Lim et al.'s TFT architecture to predict physiological residuals over a UKF prior.

Category 4: Data, Cohort, and Splits

Q14 How many total subjects were used in developing the synthetic model?

Direct Spoken Answer: The synthetic cohort contains exactly 293 subjects. These were split deterministically into 205 training subjects, 44 validation subjects, and 44 sealed test subjects. The live application serves 249 of these subjects (the training and validation sets).

> ***One-Sentence:*** *293 total synthetic subjects: 205 train, 44 validation, and 44 sealed test subjects.*

`Source: runs/pr_tft_ablation_arm_d/split.json & docs/submission/CLAIM_EVIDENCE_LEDGER.md C12`

Likely Follow-Up: Why are only 249 subjects served if there are 293 total?

Because the 44 test subjects are quarantined to prevent any leakage or accidental demonstration of test outcomes; querying them returns HTTP 404.

⚠️ What NOT to Claim: Do not claim the 249 served subjects are all type 2 diabetes patients; the cohort includes 88 healthy, 107 prediabetic, and 54 type 2 individuals.

Q15 How many real patients were used in the locked evaluation?

Direct Spoken Answer: Our nominal real-patient pool comprised 161 patients across CGMacros, BIG IDEAs, and ShanghaiT2DM. However, ShanghaiT2DM continuous glucose data was sampled at native 15-minute intervals, which violates our preregistered 5-minute coverage rule, so its test partition was honestly declared Not Estimable. The final evaluated test set consists of exactly 11 held-out test patients with native 5-minute CGM.

> ***One-Sentence:*** *11 held-out real test patients with 5-minute CGM, drawn from a nominal pool of 161 participants across three open clinical repositories.*

`Source: docs/evidence/real_data/results.json & docs/submission/LOCKED_RESULTS_RECONCILIATION.md`

Likely Follow-Up: Isn't 11 patients a very small test cohort?

Yes, it is a key limitation. Open-access continuous glucose datasets with native 5-minute sampling in type 2 diabetes are scarce. This is why we preregistered a future confirmation protocol on the 944-patient AI-READI cohort.

⚠️ What NOT to Claim: Never claim to have evaluated hundreds of real patients on the held-out test set. Always cite the exact denominator: 11 patients.

Category 5: Results and Baselines

Q17 What are your headline accuracy results on synthetic data?

Direct Spoken Answer: On our 44 synthetic validation patients, the PR-TFT hybrid model achieved a 60-minute RMSE of 19.53 mg/dL. This represents a 16.0% error reduction compared to the biophysical twin alone (23.23 mg/dL) and a 45.8% reduction compared to persistence (36.05 mg/dL).

> ***One-Sentence:*** *60-minute RMSE of 19.53 mg/dL, beating the physics twin by 16.0% and persistence by 45.8%.*

`Source: runs/pr_tft_ablation_arm_d/run_manifest.json & docs/evidence/validation_ceiling_results.json`

Likely Follow-Up: How did it perform on the type 2 diabetes subgroup specifically?

On the 10 type 2 diabetes validation patients, the 60-minute RMSE was 19.18 mg/dL, an improvement of 4.49 mg/dL over the physics twin (23.67 mg/dL).

⚠️ What NOT to Claim: Do not present synthetic validation numbers as proof of real-world human accuracy.

Q18 What are your headline accuracy results on real patients?

Direct Spoken Answer: In our locked real-patient test on 11 held-out patients, our hybrid model achieved a 60-minute RMSE point estimate of 23.11 mg/dL (95% CI 18.75 to 28.12). This outperformed the persistence baseline at 26.27 mg/dL and the reduced twin prior at 24.73 mg/dL.

> ***One-Sentence:*** *60-minute RMSE of 23.11 mg/dL on held-out real test patients, beating persistence (26.27) and physics alone (24.73).*

`Source: docs/evidence/real_data/results.json & docs/submission/LOCKED_RESULTS_RECONCILIATION.md`

Likely Follow-Up: Why does RESULTS.md say 23.04 mg/dL while results.json says 23.11 mg/dL?

As documented in our reconciliation ledger, 23.11 is the exact point estimate on the test set, while 23.04 is the mean of 2,000 bootstrap resamples from the exact same run. Both come from the same execution.

⚠️ What NOT to Claim: Do not claim to beat competitors with self-reported 20.1 mg/dL RMSE, because their evaluation did not enforce strict 1-to-1 alert matching.

Category 6: Negative Findings and Scientific Integrity

Q21 What is Gate 3, and why did it fail?

Direct Spoken Answer: Gate 3 was a preregistered synthetic realism audit. We trained a 40-feature random forest discriminator to see if it could tell our synthetic CGM traces apart from an independent literature-standard mock-real generator. The acceptance threshold was AUROC ≤ 0.65. The classifier scored AUROC 0.932, so Gate 3 failed. The top distinguishing feature was trajectory smoothness; our ODEs were smoother than real sensor data.

> ***One-Sentence:*** *A synthetic realism classifier easily distinguished our synthetic data from mock-real data (AUROC 0.932 vs ≤0.65), primarily because our ODE curves were too smooth.*

`Source: backend/synthetic_v2/validation/validation_report.json check G3.1 & docs/presentation/presentation.html Slide 12`

Likely Follow-Up: Does this mean your model was trained on fake, useless data?

No; it proves our synthetic simulator captures broad metabolic kinetics but lacks micro-level sensor noise and erratic behavioral fluctuations. That is precisely why we subsequently validated the model on real clinical datasets.

⚠️ What NOT to Claim: Never claim that Gate 3 passed or was an unimportant check.

Q24 What is the single-step jump artefact, and how do you handle it?

Direct Spoken Answer: In approximately 19.8% of sampled synthetic validation forecasts, the PR-TFT median trajectory jumped by more than 20 mg/dL in a single 5-minute step. This is an unphysical machine learning artefact. We developed a 3-point median filter that reduces jumps to 2.3%, but rather than silently altering our model weights post-hoc, our live UI flags these jumps with a visible circled indicator.

> ***One-Sentence:*** *In ~20% of forecasts, the neural network exhibits an unphysical >20 mg/dL 5-minute jump; we openly flag these blips on the chart rather than hiding them.*

`Source: docs/evidence/jump_experiment/RESULTS.md & docs/demo/JURY_BRIEF.md §5`

Likely Follow-Up: Why didn't you just retrain the model with a smoothness penalty?

Retraining would have violated our locked evaluation protocol. In science, you report the performance of the model you actually evaluated rather than silently swapping checkpoints. Smoothness regularization is slated for our next retraining cycle.

⚠️ What NOT to Claim: Do not claim the single-step jump is a real physiological phenomenon; it is an unphysical model artefact.

Category 12: Difficult and Adversarial Questions

Q35 "Your 30-minute sensitivity on real patients is only 6.9%. Isn't a model that misses 93% of spikes practically useless?"

Direct Spoken Answer: On that specific locked real-patient evaluation, 6.9% is indeed modest, and we report it honestly. However, context is vital: to avoid data snooping, meal logs were masked to zero because retrospective logs lacked timing verification. Furthermore, our model kept false alerts down to 0.11 per day with 65.5% precision. The linear trend baseline caught 44% of spikes, but spammed patients with nearly 4 false alarms every day. Catching fewer spikes with high precision is clinically preferable to alert fatigue that causes patients to abandon the monitor.

> ***One-Sentence:*** *6.9% reflects conservative operation with masked meal logs, prioritizing zero alert fatigue (0.11 false alarms/day) over noisy alarms.*

`Source: docs/evidence/real_data/results.json & docs/jury/JURY_PREPARATION_GUIDE.md §8`

Likely Follow-Up: How do you plan to raise that sensitivity to a clinically acceptable level?

In our exploratory C2 candidate, combining a risk-gated trend policy with worst-fold threshold selection increased sensitivity to 27.3% while holding false alerts to 0.87 per day. We have preregistered this policy for formal confirmation on the AI-READI cohort.

⚠️ What NOT to Claim: Do not argue that 6.9% sensitivity is already sufficient for clinical adoption.

Q38 "Why didn't you evaluate your model on AI-READI before submitting?"

Direct Spoken Answer: AI-READI CGM data resides on the FAIRhub repository under controlled access. Gaining access requires institutional CILogon authentication, verified research ethics certification, and formal data use agreements. We have written, committed, and frozen the evaluation harness (`aireadi_evaluate.py`), but in accordance with ethics laws, we cannot execute it until formal data access is granted.

> ***One-Sentence:*** *AI-READI requires institutional ethics and data-access approval; our protocol is frozen in Git awaiting access authorization.*

`Source: docs/evidence/aireadi_confirmation/PREREGISTRATION.md`

Likely Follow-Up: What if AI-READI data disproves your candidate policy?

Then we will publish that negative result with the exact same transparency that we published our Gate 3, Hall, and AZT1D results.

⚠️ What NOT to Claim: Never claim that AI-READI evaluation is already underway or partially completed.

Q39 "What would you do if a judge asks for a metric or clinical subgroup that you haven't measured?"

Direct Spoken Answer: We will say: *"We have not established that yet. Our current evidence demonstrates [mention closest verified metric], but to answer your question properly, our next step would be [specific defensible experiment]."* We will never invent numbers on the spot.

> ***One-Sentence:*** *State plainly that it has not been measured yet, cite the verified evidence, and outline the exact experiment needed.*

`Source: Master instructions & docs/jury/JURY_PREPARATION_GUIDE.md §1`

Likely Follow-Up: Can you estimate what the number would probably be?

In clinical research, guessing unmeasured metrics can lead to false safety assumptions. We prefer to state our known evidence bounds and test the metric empirically.

> ⚠️ What NOT to Claim: Never guess or bluff a number.